Jiawen (Jarvan) Wang

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PROFILE

Master of Information Technology student at the University of Technology Sydney, with graduation expected in December 2026. Practical experience in Python, SQL, data analysis, and machine learning through projects in sports analytics, computer vision, NLP, and healthcare prediction.

CORE STRENGTHS

- Builds end-to-end workflows for data preparation, analysis, feature engineering, modeling, visualization, and error review.
- Evaluates models with Accuracy, F1-score, ROC-AUC, confusion matrices, validation curves, and cross-validation.
- Uses Python, SQL, Excel, Tableau, Scikit-learn, PyTorch, and Jupyter/Colab notebooks for reproducible analysis and experiments.

EDUCATION

University of Technology Sydney 2025-Present

Master of Information Technology | Selected Coursework: Deep Learning, Natural Language Processing Algorithms, Advanced Data Analytics, Data Visualisation and Visual Analytics, Research Project

Huazhong Agricultural University 2019-2023

Bachelor of Biotechnology | Coursework: Microbiology, Cell Biology, Genetics, Biochemistry, Genetic Engineering, Pharmacology

PROJECTS

T20 Cricket Analytics & AI Prototype

Python, Scikit-learn, XGBoost, React, TypeScript, LLM API

- Built a 10-stage, leakage-aware research pipeline across **1,521 T20I matches** and **360,721 events**, comparing four model families with chronological validation, probability calibration, and paired match-level bootstrap evaluation.
- Developed a React/TypeScript analytics prototype for match replay, win-probability and player-contribution exploration, integrating an LLM-powered Q&A assistant for conversational analysis.

Deep Learning for Image Classification and Object Detection

PyTorch, ResNet18, Faster R-CNN, SSD300

- Built PyTorch pipelines for image classification and object detection with **ResNet18, Faster R-CNN, and SSD300**, using annotations and predictions for model debugging.
- Compared baseline and customized ResNet18 architectures using training/validation loss and accuracy curves to assess convergence and overfitting.

Handwritten Mathematical Symbol Recognition

Python, Scikit-learn, OpenCV, HOG, LBP

- Engineered HOG, LBP, and raw-pixel feature representations for a multi-class symbol recognition workflow.
- Trained SVM and ANN classifiers; compared confusion matrices and accuracy metrics to identify misclassification patterns.

Cardiovascular Disease Prediction

Python, Scikit-learn, Pandas, NumPy

- Cleaned and engineered features from **56,000+ medical records**, including missing-value handling and structured preprocessing.
- Built Logistic Regression, Random Forest, and KNN models; compared Accuracy, F1-score, and ROC-AUC to select the strongest model.

TECHNICAL SKILLS

Analytics & BI: Excel, Tableau, SQL, Pandas, NumPy, Matplotlib, Seaborn, data cleaning, visualization, reporting

Machine Learning & AI: Scikit-learn, PyTorch, XGBoost, Random Forest, CNN, model evaluation, probability calibration, cross-validation

Programming, Cloud & Tools: Python, React, TypeScript, Linux/Unix, Git/GitHub, Jupyter/Colab, VS Code, OpenCV, LLM API integration, basic AWS exposure | English, Mandarin Chinese